DAY_01

DURATION : 2hrs

1. Rebuild résumé form using Tailwind CSS utility classes
2. Make the layout fully responsive using Tailwind breakpoints

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DAY_02

DURATION : 2hrs

1. Practice writing basic JavaScript programs for better understanding of the language constructs
2. ES6+ based exercises

DAY_03

DURATION : 2hrs

1. Sort a list of elements by implementing a sorting algorithm.
2. Write a JavaScript program to list the properties of a JavaScript object.
3. Rewrite a prototype-based object using ES6 class syntax
4. Functional programming exercises: transform a dataset using map/filter/reduce

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DAY_04

DURATION : 2hrs

1. Write a JavaScript function to get First and Last name from the previously created Resume form
2. Validate the entire Resume form using client-side JavaScript
3. Write a JavaScript function to validate whether a given value is RegEx or not.
4. Fetch data from a JSON endpoint using Fetch API and display results with error handling

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DAY_05

DURATION : 2hrs

1. Setup a TpeScript project and convert a plain JS file to TS and fix all type errors
2. Define interfaces for an API response, type a Fetch/Axios call end-to-end
3. Build a typed DOM utility module, write a generic function used in 2+ contexts
4. Convert the resume from JS to TS

DAY_06

DURATION : 2hrs

1. Live news headlines using Axios with typed TypeScript API response

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DAY_07 and DAY_08

DURATION : 2hrs

1. Assignment on JavaScript callback functions
2. Assignment on Timers, Promises, and Async& Await
3. Implement Promise.all to fetch multiple endpoints concurrently and handle partial failures

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DAY_09

DURATION : 2hrs

1. Use Node and Express to write a simple web application that consists of at least 5 route implementations
2. Rebuild any previous Node assignment using Express and a template engine

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DAY_10

DURATION : 2hrs

1. Rebuild any previous plain HTML lab assignment using React functional components
2. Build a React Clock app showing time (hh:mm:ss) of any three countries

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DAY_11 and DAY_12

DURATION : 4hrs

1. Update Clock app time using useState and useEffect
2. Implement the following items in the React Clock app o Update the time (hh:mm:ss) using State and Lifecycle methods o Add a close functionon each rendered clock component o Assign background color of rendered clock components based on AM, PM

DAY_13 and DAY_14

DURATION : 4hrs

1. Implement and integrate a new feature in the React Clock app where one can select a country time zone from dropdown list and click on “Add” button to render it
2. Use useRef to focus an input field programmatically
3. Extract clock logic into a custom useClockTick hook
4. Demonstrate useCallback preventing unnecessary child re-renders

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DAY_15

DURATION : 2hrs

1. Implement error boundaries at appropriate places in the React Clock app
2. Add React router to the clock app
3. Implement useNavigate for programmatic navigation and useParams for dynamic routes

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